ID	Work Package	Component	Requirement name	Description	Rationale	Acceptance criteria	Parent customer spec	Priority	Risk	Approved
FR-1	DG-4	Digger	Recipe Interface	The digger exposes an interface to receive a recipe containing at least a set of desired excavation poses and the respective paths between them.	Ensures continues digging in areas without continuous connection to the FM.	Digger exposes an interface to receive a recipe.	CN-24	Should	Low	By team and customer
FR-2	DG-4	Digger	Recipe Executor	Digger executes the recipe received from the FM.	Ensures continues digging in areas without continuous connection to the FM.	Diggers execute recipe without FM communication.	CN-24	Should	Low	By team and customer
FR-61	DG-4	Digger	Recipe Confirmation	Digger confirms recipe receival	Ensures continues digging in areas without continuous connection to the FM.	Digger confirms recipe receival.	CN-24	Must	Low	By team and customer
FR-62	DG-1	Digger	Status Reporting	Digger provides health, progress and other diagnostics.	Enables communicating status information between to FM.	Digger sends status information to FM.	CN-17	Must	Low	By team and customer
FR-3	DG-4	Digger	Localization	Digger localizes itself.	Ensures autonomous operation.	Digger localizes within a defined coordinate system.	CN-7	Must	Medium	By team and customer
FR-4	DG-4	Digger	Local Obstacle Detection	Digger detects non-traversable obstacles.	Enables performing actions based on non-traversable objects.	Digger detects non-traversable object.	CN-7 CN-2	Must	Low	By team and customer
FR-5	DG-4	Digger	Local Obstacle Avoidance	Digger avoids non-traversable obstacle.	Reduces the need for human intervention enhancing efficiency and scalability.	Digger avoids non-traversable object.	CN-7 CN-15 CN-11	Must	Low	By team and customer
FR-63	DG-4	Digger	Local Excavation Avoidance	Digger avoids excavations.	Ensures previous excavation aren't destroyed.	Digger avoids excavations created by itself and/or other diggers.	CN-7	Should	Low	By team and customer
FR-6	DG-4	Digger	Local Planning/Control	Digger follows path between excavations specified in the recipe.	Scalability through autonomy.	Digger drives from excavation to excavation following the given path.	CN-7 CN-15	Must	Low	By team and customer
FR-7	DG-5	Digger	Safe Navigation	Digger prevents harm to the environment and bystanders during navigation.	Ensures safety.	In accordance with FMEA and safety regulations.	CN-7 CN-8 CN-2	Must	Medium	By team and customer
FR-8	DG-4	Digger	Navigation Recovery/Fallback Behavior	Digger executes defined fallback/recovery behavior in defined edge cases.	Enables less human intervention in ever expanding navigational edge cases thus enhancing scalability.	Digger recovers autonomously from known edge-cases.	CN-15 CN-7	Must	Low	By team and customer
FR-9	DG-2	Digger	Feed Forward (FF) Excavation Control	Digger excavates without sensor feedback on excavation quality.	Ensures autonomous operation without having to analyze sensor data regarding the excavation specifications.	Digger autonomously excavates by executing a pre-defined set of steps.	CN-7 CN-10	Must	Low	By team and customer
FR-65	DG-2	Digger	Stance Compensation	Digger uses roll, pitch, yaw information to adjust excavation.	Ensured autonomous operation and feed forward digging by correcting for angular offsets.	Digger autonomously excavates by executing a pre-defined set of steps after adjusting for stance.	CN-7 CN-10	Must	Low	By team and customer
FR-66	DG-4	Digger	Digger Respects Geofence	Digger remains inside the geofence.	Ensures safe operation and prevents harm to environment surrounding excavation site.	Digger remains inside geofence when excavations are planned right up to the boundary.	CN-2 CN-23	Must	Low	By team and customer
FR-10	DG-3	Digger	Excavation Control with Sensor Feedback	Diggers adjusts its digging action based on excavation feedback.	To improve excavations success rate.	Digger autonomously excavates by executing predefined steps and changes its behavior based on sensor data.	CN-7	Must	Low	By team and customer
FR-12	DG-5	Digger	Safe Excavation	Digger prevents harm to the environment and bystanders during excavation.	Ensures safety.	In accordance with FMEA and safety regulations.	CN-7 CN-8 CN-2	Must	Low	By team and customer
FR-13	DG-2	Digger	Excavation Recovery/Fallback Behavior	Digger executes defined fallback/recovery behavior in defined edge cases. (e.g. skipping excavation when an obstacle is detected)	Enables less human intervention in ever expanding digging edge cases thus enhancing scalability.	Digger recovers autonomously from known edge-cases.	CN-15 CN-7 CN-11	Must	Low	By team and customer
FR-11	DG-3	Digger	Excavation Fallback Behavior - Skip Excavation	The digger skips a planned excavation if an obstacle is detected at its excavation location.	Expands the range of allowable ground conditions	The digger autonomously detects obstacles at the assigned excavation location and proceeds to the digging end pose.	CN-7	Could	Low	By team and customer
FR-14	DG-2	Digger	Machine Safety	Digger stops on detected diagnostic errors (e.g. detected hardware failures)	Ensures safety and prevents further machine damage.	System stops when detecting diagnostics errors.	CN-7 CN-2	Must	Low	By team and customer

FR-15	DG-2	Digger	Digger Monitoring	Digger performs self-diagnostics, e.g. battery level.	Ensures status inspection.	Digger performs self-diagnostics.	CN-6 CN-17	Must	Low	By team and customer
FR-16	DG-4	Digger	Data Exchange Manager	Digger buffers and prioritize critical data to be send to the FM.	Ensures FM has access to data required for doing status reporting, planning, recipe creation, etc.	FM remains full overview of relevant digger data after communication outage.	CN-15 CN-24	Should	Low	By team and customer
FR-17	DG-2	Digger	Charging Pause	The digger pauses operation when a lower battery limit is observed.	To reserve battery capacity for theft protection and manual control.	Digger pauses when lower limit on battery is detected, and continues when the limit is exceeded again.	CN-18 CN-20	Must	Low	By team and customer
FR-18	DG-1	Digger	Joint state feedback	Digger exposes all relevant joint states.	Enables autonomous operation.	Digger, or appended sensors, exposes all joint states.	CN-7	Must	Low	By team and customer
FR-19	DG-2	Digger	Cease excavation upon exceeding the hydraulic pressure threshold.	Lift arm on exceeding the hydraulic pressure threshold.	Prevent damage when the bucket gets stuck, e.g. on hitting a rock whilst digging.	The preventive action reduces pressure below the threshold.	CN-2 CN-7 CN-11	Must	Low	By team and customer
FR-20	DG-2	Digger	Cease excavation upon exceeding the tilt threshold.	Lift arm on exceeding the tilt threshold.	Ensure the digger remains stable and does not lift off the ground.	The preventive action reduces the tilt below the threshold.	CN-2 CN-7 CN-11	Must	Low	By team and customer
FR-21	DG-1	Digger	Actuator control	All required digger actuators, including the tracks, are controllable via a digital interface.	Enables autonomous operation.	We can control the actuators by sending commands from a PC.	CN-7	Must	Medium	By team and customer
FR-22	DG-1	Digger	Teleoperation	The digger's actuators can be controlled by a user that has direct line-of-sight.	Enables easy set-up and user intervention.	We can manually send commands to control the actuators.	CN-22 CN-12	Must	Low	By team and customer
FR-23	DG-1	FM	Remote Software Deployment	Software components are be deployed over-the-air.	Enables easy set-up and maintenance.	Software is deployed on to digger over-the-air.	CN-12 CN-13	Could	Low	By team and customer
FR-24	FM-1	FM	Communication System	Diggers, UIs, scouts and FMs but communicate with each other.	Enables communication between devices and programs in the system.	System critical communication deemed reliable in remote areas.	CN-17 CN-7	Must	Medium	By team and customer
FR-25	FM-3	FM	Satellite Mapping	FM collects satellite imagery and height information about a region of interest.	Enable site-manager to geo-fence region of interest. Enables planning of excavations and scout navigation based on elevation information.	Satellite data obtainable for any random area.	CN-19	Must	Medium	By team and customer
FR-26	SC-1	FM	Scout Navigation	Scout collects data over geotagged areas autonomously.	Enables faster more efficient and more accurate data collection, reducing the need for recollection and manual teleoperation, thus enhancing system scalability.	Path created for scout navigation automatically.	CN-7 CN-15 CN-19	Must	Low	By team and customer
FR-27	FM-3	FM	Excavation Planning	FM plans excavations over the feasible area within the geofenced region automatically.	Enables faster and more efficient excavation planning requiring less to no manual input, thus enhancing scalability.	Excavations planned automatically over the feasibility area.	CN-7 CN-15 CN-11	Must	Low	By team and customer
FR-28		FM	Excavation Allocation	FM dynamically re-allocates excavations to other diggers when a digger becomes unavailable.	Dynamic reallocation will firstly complicate planning and secondly might be undesired since driving to another digger excavation location might be inefficient.			Won't	Low	By team and customer
FR-29	FM-2	FM	Digger Path Planning	FM plans the paths between excavations for the diggers	Allows for central route optimization, and future incorporation of environment data.	FM plans paths between excavations.	CN-11	Must	Low	By team and customer
FR-30	FM-2	FM	Recipe Generator	FM generates recipes for the diggers.	Enables digging with less FM supervision enhancing robustness.	Diggers execute recipe without FM communication.	CN-24	Should	Low	By team and customer
FR-31	FM-4	FM	Weather Forecasting	FM predicts wind risks.	Enables system to take action reducing potential wind damage.	Weather triggers defensive positioning timely.	CN-18	Must	Medium	By team and customer
FR-32	FM-4	FM	Recipe Interuption	FM interrupts active diggers to take preventive actions against wind damage	Enables system to take action in case of wind risk reducing potential wind damage.		CN-18	Must	Low	By team and customer
FR-33	SC-1	FM	Scout Mapping	Automatically convert scout footage into a map.	Enables faster and more efficient map creation requiring less to no manual input, thus enhancing system scalability.	Map generated from scout data automatically.	CN-15 CN-19 CN-11	Must	Low	By team and customer

FR-34	FM-3	FM	Autonomous Terrain Analysis	FM identifies feasible terrain automatically.	Enables faster a more efficient deployment requiring less to no manual input, thus enhancing scalability and efficiency.	Automatic terrain classification.	CN-15 CN-19 CN-11	Must	Low	By team and customer
FR-35	FM-1	FM	Diagnostics API	FM provides an interface for devices to send and/or receive diagnostics data.	Enables remote monitoring.	FM sends and receives status information.	CN-17 CN-13	Must	Low	By team and customer
FR-36	DG-5	FM	Media API	FM provides an interface for devices to send and/or receive media.	Enables remote monitoring and viewing for anti-theft prevention.	FM sends and receives media data.	CN-17 CN-20	Must	Medium	By team and customer
FR-37	FM-1	FM	Diagnostics Database	FM stores health, progress and other diagnostics into a database.	Enables data recollection.	FM stores data into database.	CN-17	Must	Low	By team and customer
FR-38	FM-1	FM	Status API	FM provides an interface for devices to send and/or receive status information.	Enables communicating status information between the FM and devices.	FM sends and receives status information.	CN-17	Must	Low	By team and customer
FR-39	FM-1	FM	Status Reporting	FM consolidates diagnostics and progress data from database into status information.	Enables users to easily draw conclusion on status and progress.	FM consolidates database data into a status report.	CN-17 CN-13	Could	Low	By team and customer
FR-40	FM-5	FM	Site Management API	FM provides an interface to view/edit site information to regional and general managers.	Enables editing and viewing sites.	Regional/general manager adds/removes/edit sites.	CN-17 CN-15	Could	Low	By team and customer
FR-41	FM-5	FM	User Permission Management API	FM provides an interface for devices to set permissions.	Enhances safety, accountability and reduces human error.	Each class of user sets/gets user permissions according to class description.	CN-2 CN-17	Could	Low	By team and customer
FR-42	FM-5	FM	Device Management API	FM provides an interface to add, remove and edit devices belonging to a respective site.	Ensures easy setup and ease of use.	User adds/removes/edits devices such as diggers.	CN-12	Must	Low	By team and customer
FR-43	FM-2	FM	Geofencing API	FM provides an interface for users to geofence a region of interest.	Enables efficient and easy setup.	User adds/removes/edits geofence.	CN-19 CN-12	Must	Low	By team and customer
FR-44	FM-2	FM	Excavation Planning API	FM provides an interface for users to add, remove and edit excavation positions.	Enables efficient and easy setup.	FM allows manual excvation planning	CN-12 CN-19	Must	Low	By team and customer
FR-45	FM-3	FM	Feasible Terrain API	FM provides an interface to add, remove, edit feasible terrain regions.	Enables safe operation by excluding regions that should not be traversed nor excavated.	User adds/removes/edits feasible terrain.	CN-23 CN-2 CN-12	Could	Low	By team and customer
FR-46	DG-5	FM	Media Database	FM stores security footage, scout footage and other media.	Enables reviewing security footage.	FM stores media data in database.	CN-20 CN-17	Should	Low	By team and customer
FR-47	FM-2	FM	Recipe API	The FM interface exposes a recipe interface.	Enables digging with less FM supervision enhancing robustness.	Device requests and receives recipe from FM.	CN-24	Must	Low	By team and customer
FR-48	UI-1	UI	Status Reporting	GUI provides status report interface to site manager.	Enables status reviewing.	Site manager views correct system status from a GUI.	CN-17	Could	Low	By team and customer
FR-49	FM-5	UI	Site Editor	Users can manage multiple sites.	Enables efficient access to multiple sites and respective site data enhancing scalability.	Regional/general manager adds/removes/edit sites.	CN-17 CN-15	Could	Low	By team and customer
FR-50	FM-5	UI	User Permission Editor	Users can edit permission settings according to user class (i.e. operator, site manager, etc.)	Enhances safety, accountability and reduces human error.	Each class of user sets/gets user permissions according to class description.	CN-2 CN-1	Could	Low	By team and customer
FR-51	FM-5	UI	Device Editor	Users can add, remove and edit devices belonging to the respective site.	Ensures easy setup and ease of use.	User adds/removes/edits devices such as diggers.	CN-12	Must	Low	By team and customer
FR-52	UI-2	UI	Geofencing Editor	Users can geofence a region and denote specific exclusion regions.	Enables efficient and easy setup.	UI allows geofencing.	CN-19 CN-12	Must	Low	By team and customer
FR-53	UI-2	UI	Excavation Editor	Users can manually add, remove and edit desired excavation positions.	Enables efficient and easy setup.	UI allows manual excavation planning	CN-12 CN-19	Must	Low	By team and customer
FR-54	UI-2	UI	Feasible Terrain Editor	Users can add, remove, edit feasible terrain regions.	Enables safe operation by excluding regions that should not be traversed nor excavated.	User adds/removes/edits feasible terrain.	CN-1 CN-2	Must	Low	By team and customer
FR-55	UI-3	UI	Teleoperator	Users can teleoperate the digger while staying within direct line of sight.	Enables easy setup and human intervention for efficient recovery, thus enhancing scalability.	User can control both the tracks as the arm of the digger while being in direct line of sight.	CN-12 CN-22 CN-15	Must	Low	By team and customer
FR-56	ST-1	Solar	Trailer Mechanical Interface	The trailer mechanically interfaces with the digger.			CN-6 CN-9	Must	Low	By team and customer
FR-57	ST-1	Solar	Trailer Modularity	The trailer is modular, allowing it to be disassembled.	For more economical packing during shipping.		CN-6 CN-9	Must	Low	By team and customer

FR-58	ST-2	Solar	Trailer Digital Interface	The trailer digitally interfaces with the digger.			CN-6 CN-9	Could	Low	By team and customer
FR-59	ST-1	Solar	Trailer Electrical Interface	The trailer electrically interfaces with the digger.			CN-6 CN-9	Must	Low	By team and customer
FR-67	ST-1	Solar	Solar Power Generation	The trailer generates solar power			CN-9	Must	Low	By team and customer
FR-68	FM-1	FM	Robot configuration management	The FM must manage robot specific configurations	Configuration management is necessary for robot specific configurations such as calibration values.		CN-12 CN-15	Must	Low	By team and customer
FR-60	SC-1	Scout	Scout Docking	Scout autonomously docks.	Enables faster, more frequent and efficient deployment requiring less to no user input, thus enhancing system scalability and increasing scouting, which could improve efficiency by more accurate mapping	Autonomous docking off the scout.	CN-15 CN-19	Could	Low	By team and customer
FR-69	DG-1	Digger	Remote shutdown	Digger exposes an interface to be shut-down.	To disable non-functioning robots			Must	Low	By team and customer
FR-70	DG-4	Digger	Bund water capture	Any bund should catch the water that flows inbetween the above bunds		On a non-curved slope the gradient of the slope will always be crossed by a bund within 2 bund-rowlines	CN-6 CN-9	Must	Low	By team and customer
								Must	Low	No